



Circular Palindromes

Problem Submissions Leaderboard Discussions

A palindrome is a string that reads the same from left to right as it does from right to left.

Given a string, S , of N lowercase English letters, we define a k -length rotation as cutting the first k characters from the beginning of S and appending them to the end of S . For each S , there are N possible k -length rotations (where $0 \leq k < N$). See the Explanation section for examples.

Given N and S , find all N k -length rotations of S ; for each rotated string, S_k , print the maximum possible length of any palindromic substring of S_k on a new line.

Input Format

The first line contains an integer, N (the length of S).
The second line contains a single string, S .

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $0 \leq k < N$
- S is comprised of lowercase English letters.

Output Format

There should be N lines of output, where each line k contains an integer denoting the maximum length of any palindromic substring of rotation S_k .

Sample Input 0

```
13
aaaaabbbbaaaa
```

Sample Output 0

```
12
12
10
8
8
9
11
13
11
9
8
8
10
```

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Difficulty **Advanced**
Max Score 120
Submitted By 13/747

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```
1  import java.util.*;
2
3  public class Main {
4      static int longestPal(String s) {
5          int max = 1;
6
7          for (int i = 0; i < s.length(); i++) {
8              int l = i, r = i;
9
10             while (l >= 0 && r < s.length() &&
11                 s.charAt(l) == s.charAt(r)) {
12                 max = Math.max(max, r - l + 1);
13                 l--;
14                 r++;
15             }
16
17             l = i;
18             r = i + 1;
19
20             while (l >= 0 && r < s.length() &&
21                 s.charAt(l) == s.charAt(r)) {
22                 max = Math.max(max, r - l + 1);
23                 l--;
24                 r++;
25             }
26         }
27         return max;
28     }
29
30     public static void main(String[] args) {
31         Scanner sc = new Scanner(System.in);
32         int n = sc.nextInt();
33         String s = sc.next();
34
35         for (int i = 0; i < n; i++) {
36             String rotation = s.substring(i) + s.substring(0, i);
37             System.out.println(longestPal(rotation));
38         }
39     }
40 }
```

Line: 48 Col: 2

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

✓ Sample Test case 1

✓ Sample Test case 2

Input (stdin)

```
1 13
2 aaaaabbbbbaaaa
```

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Your Output (stdout)

```
1 12
2 12
3 10
4 8
5 8
6 9
7 11
```